



Section-A

- Q.1) (B) Calcium and Magnesium
- Q.2) (A) $\text{Mg} : \text{O} : \rightarrow \text{Mg}^{2+} [\text{O}^{2-}]$
- Q.3) (C) It has weak electrostatic forces of attraction between its oppositely charged ions
- Q.4) (A) Salt and water is formed
- Q.5) (B) 5
- Q.6) (B) Al_2O_3 and MgO
- Q.7) (D) 1:8
- Q.8) (D) Cytoplasm and Oxygen deficient muscle cells



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Question Number

Q.9) ☒ (c) 100% round and yellow

Q.10) ☒ (c) (i) and (iii)

Q.11) ☒ (d) Auxins

Q.12) ☒ (c) Starch into simple sugars

Q.13) ☒ (d) 99 %

Q.14) ☒ (d) (ii) and (iv)

Q.15) ☒ (A) scattering of light

Q.16) ☒ (A) (i) and (ii)

Q.17) ☒ (B) Both Assertion(A) and Reason(R) are true, but Reason(R) is not the correct explanation of(A).



Q.18) (A) Both Assertion(A) and Reason(R) are true and Reason(R) is the correct explanation of Assertion(A)

Q.19) (C) Assertion(A) is true, but Reason(R) is false.

Q.20) (B) Both Assertion(A) and Reason(R) are true, but Reason(R) is not the correct explanation of Assertion(A)

Section-B

Q.21) Given -

$u = -10 \text{ cm}$ (object distance)

$f = -15 \text{ cm}$ (Since it is a concave mirror)

To find $\rightarrow v$ (image distance)

$\Rightarrow$ Mirror formula $= \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$



$$\Rightarrow \frac{1}{v} - \frac{1}{10} = -\frac{1}{15}$$

$$\Rightarrow \frac{1}{v} = -\frac{1}{15} + \frac{1}{10}$$

$$\Rightarrow \frac{1}{v} = \frac{-10 + 15}{10 \times 15}$$

$$\Rightarrow \frac{1}{v} = \frac{5}{10 \times 15 \times 2}$$

$$\boxed{v = +30\text{cm}}$$

Position of image formed by mirror = +30cm
on the other side of mirror.

Q22)

(a)

Lamp A →

Power = 50W

Volt = 220V



We know,

$$P = \frac{V^2}{R}$$

$$R = \frac{V^2}{P}$$

$$R_A = \frac{44}{\cancel{220} \times \cancel{220}} \times 220$$

$$= 44 \times 22$$

$$= \underline{968 \Omega}$$

Lamp B →

$$\text{Power} = 25 \text{ W}$$

$$\text{Volt} = 220 \text{ V}$$

$$P = \frac{V^2}{R}$$

$$R_B = \frac{44}{\cancel{220} \times \cancel{220}} \times 220$$

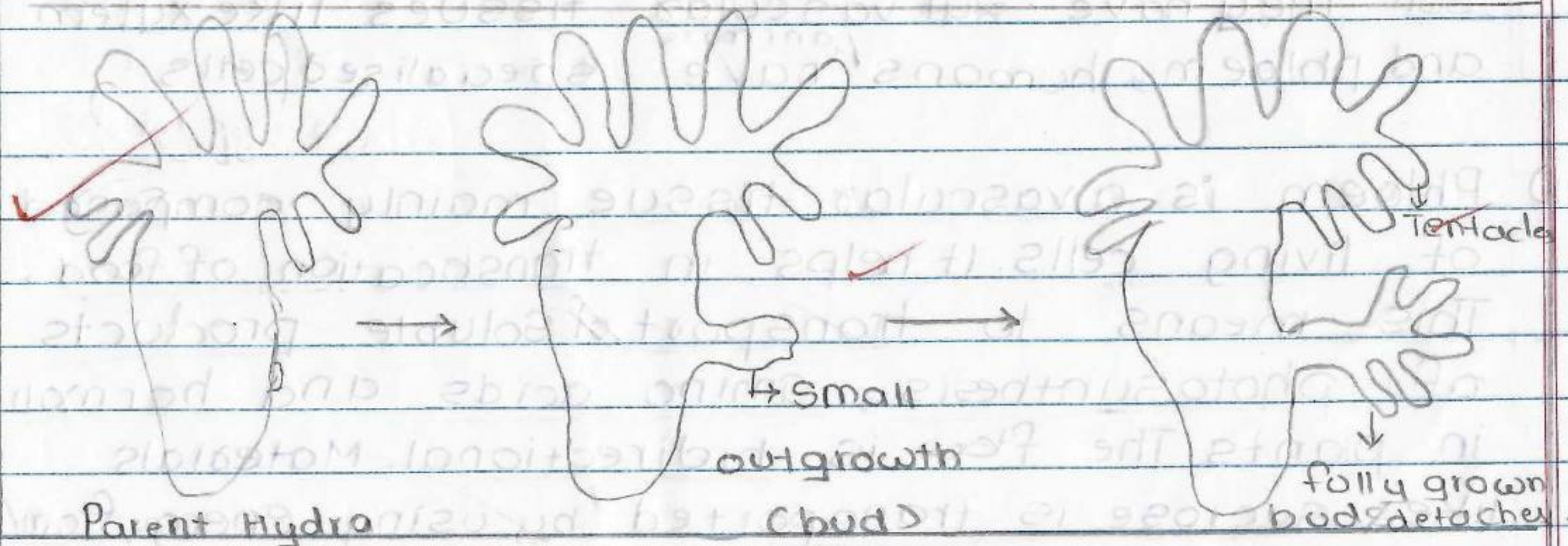
$$= \underline{1936 \Omega}$$



$$\therefore \frac{R_A}{R_B} = \frac{968}{1936}$$

$$R_A : R_B = \boxed{1 : 2} \quad \checkmark$$

Q.23) Hydra reproduces by the process of budding.
With help of regenerative cells, there is
outgrowth called bud which develops.



Diag^m - Budding in Hydra



Q.24)

(b)(i)

The transport system in plants is relatively slower than in animals because plants are mainly composed of dead cells. Plants are stationary and don't require energy to perform activities when compared to ^{of animals} humans. Moreover, simple processes like diffusion and osmosis help in transport of materials from soil. They have ~~not~~ ^{animals} vascular tissues like xylem and phloem, humans have specialised cells

(ii)

Phloem is a vascular tissue mainly composed of living cells. It helps in translocation of food. This means to transport soluble products of photosynthesis, amino acids and hormones in plants. The flow is bidirectional. Materials like sucrose is transported by using energy from ATP for eg. During spring, food stored in stem



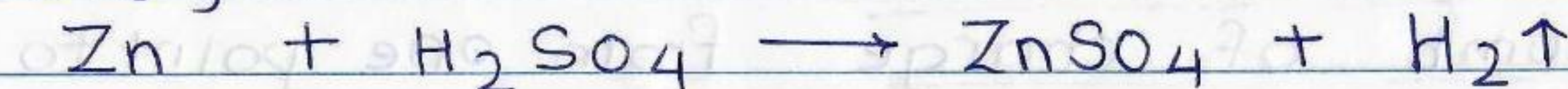
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and roots is transported to buds.

Q.25) For a chemical change to occur there has to be

- i) evolution of gas
- ii) change in temperature
- iii) change in odour
- iv) formation of precipitate

Here,

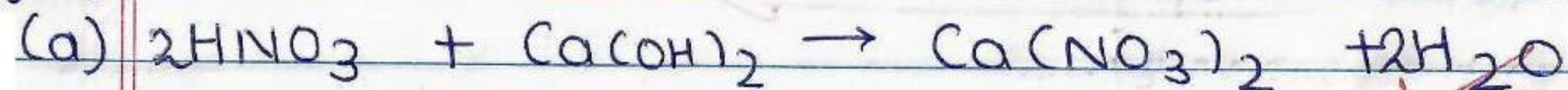


Here 2 observations are seen -

- i) Evolution of gas (H_2)
- ii) Change in temperature since this is an exothermic reaction.



Q.26)



Section-C

Q.27)

(a) When 1 Joule of work is done to move 1 Coloumb of charge from one point to another in an electric field, it is said to be 1 Volt

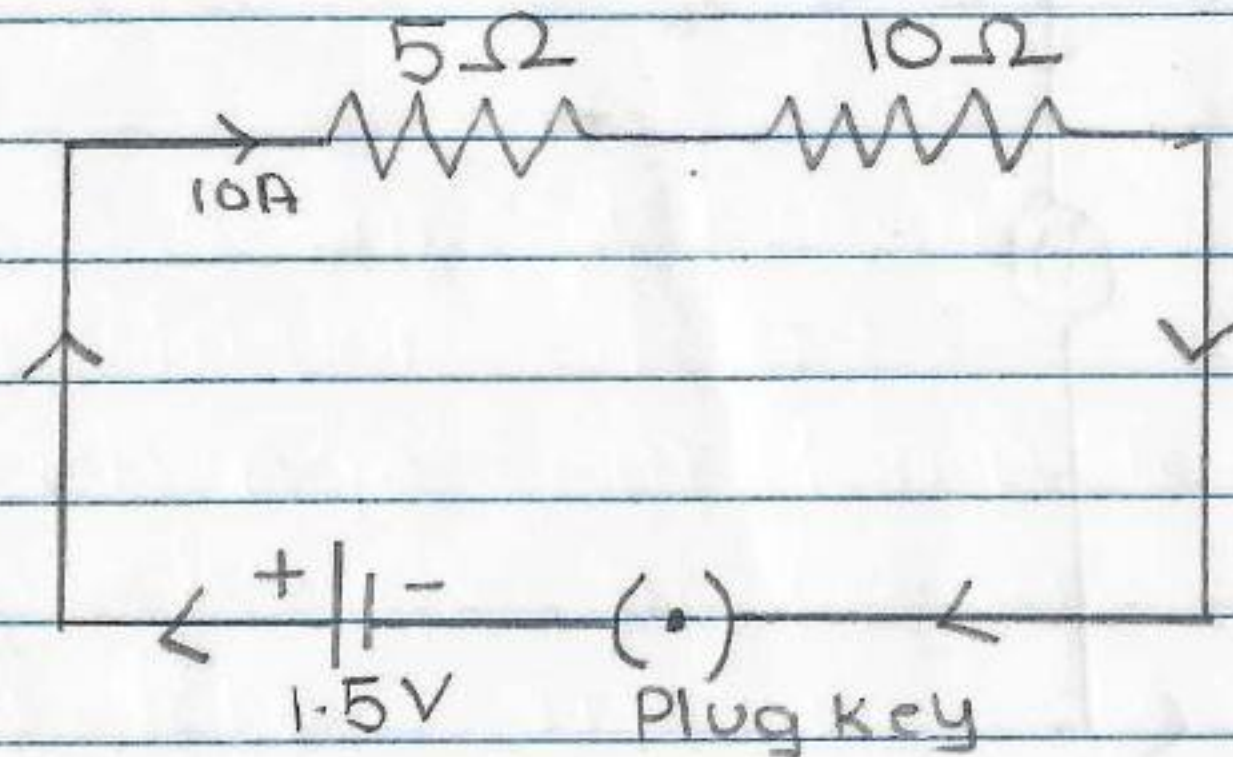
$$V = \frac{W}{Q}$$

$$1V = \frac{1J}{1C}$$

$$1V = 1JC^{-1}$$



Q.27)
(b)



Let $R_1 = 5\Omega$

$R_2 = 10\Omega$

To find $\rightarrow I$ (current)

Total resistance (R_s) = $R_1 + R_2$
 $= 5 + 10$
 $= 15\Omega$ ✓

$I = \frac{V}{R}$ (Ohm's law)

$= \frac{1.5}{15} = \frac{10A}{15} = \underline{\underline{0.1A}}$ ✓